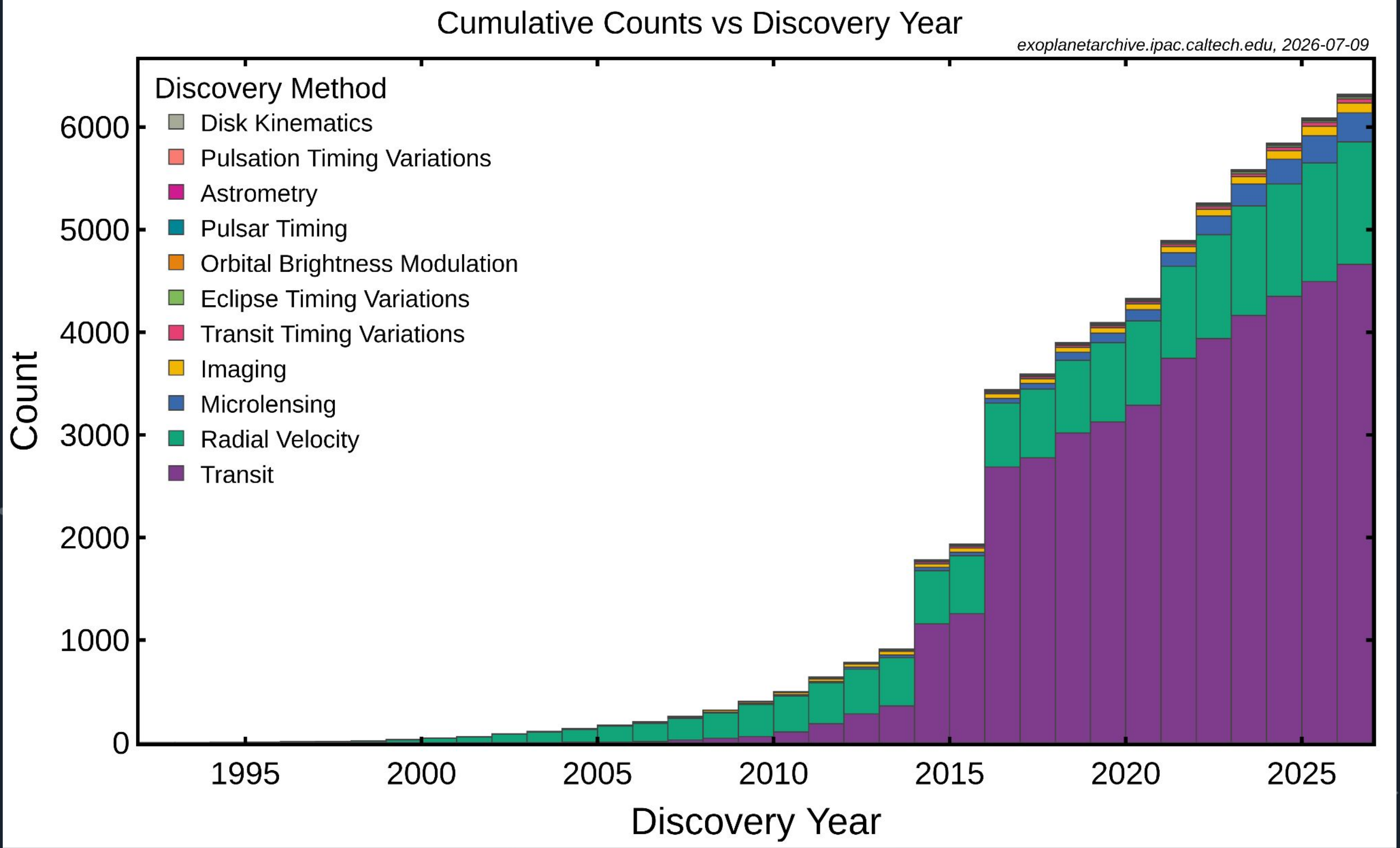
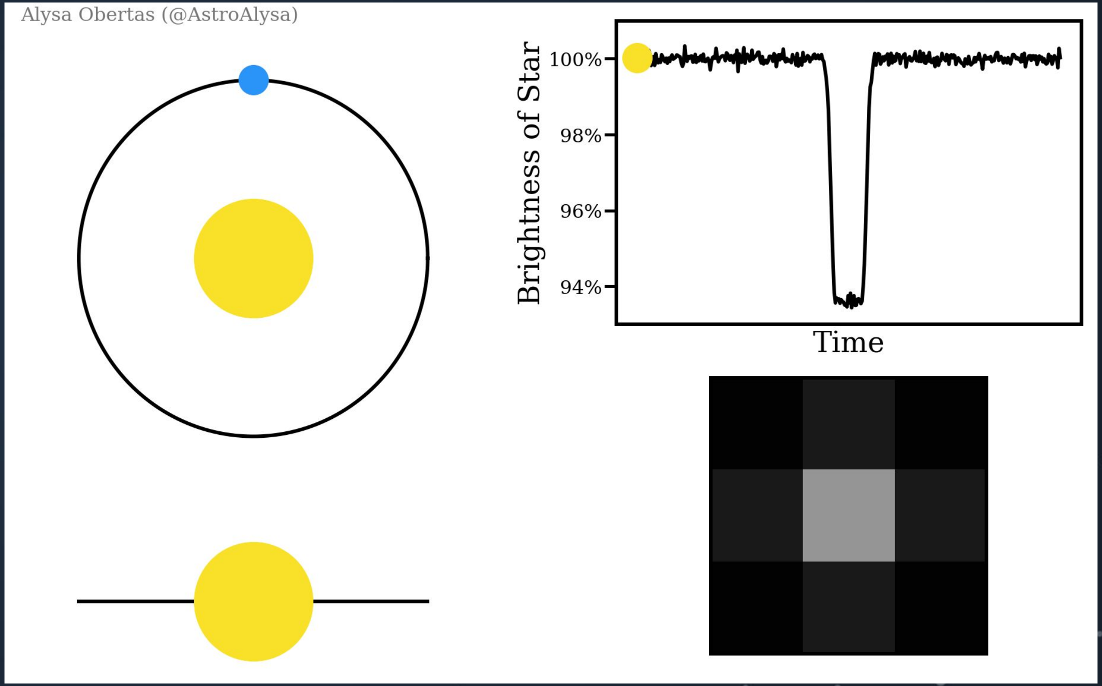
EXOPLANET DETECTION METHODS


Cumulative Counts vs Discovery Year
exoplanetarchive.ipac.caltech.edu, 2026-07-09
Discovery Method
Disk Kinematics
Pulsation Timing Variations
Astrometry
Pulsar Timing
Orbital Brightness Modulation
Eclipse Timing Variations
Transit Timing Variations
Imaging
Microlensing
Radial Velocity
Transit
Count
0
1000
2000
3000
4000
5000
6000
1995
2000
2005
2010
2015
2020
2025
Discovery Year

METHOD I: THE TRANSIT METHOD

- We can see planets **transit** their stars
- This makes the stars appear fainter for a time!
- We can see this by looking at **light curves** -- time-resolved photometry measurements

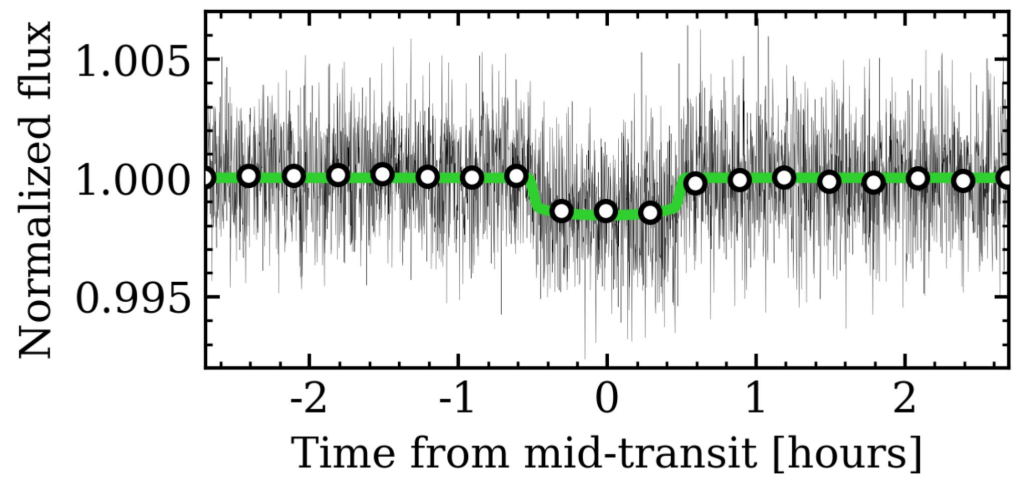


METHOD I: THE TRANSIT METHOD

- **Transit depth:**

  $$Z = (R_p/R_*)^2$$

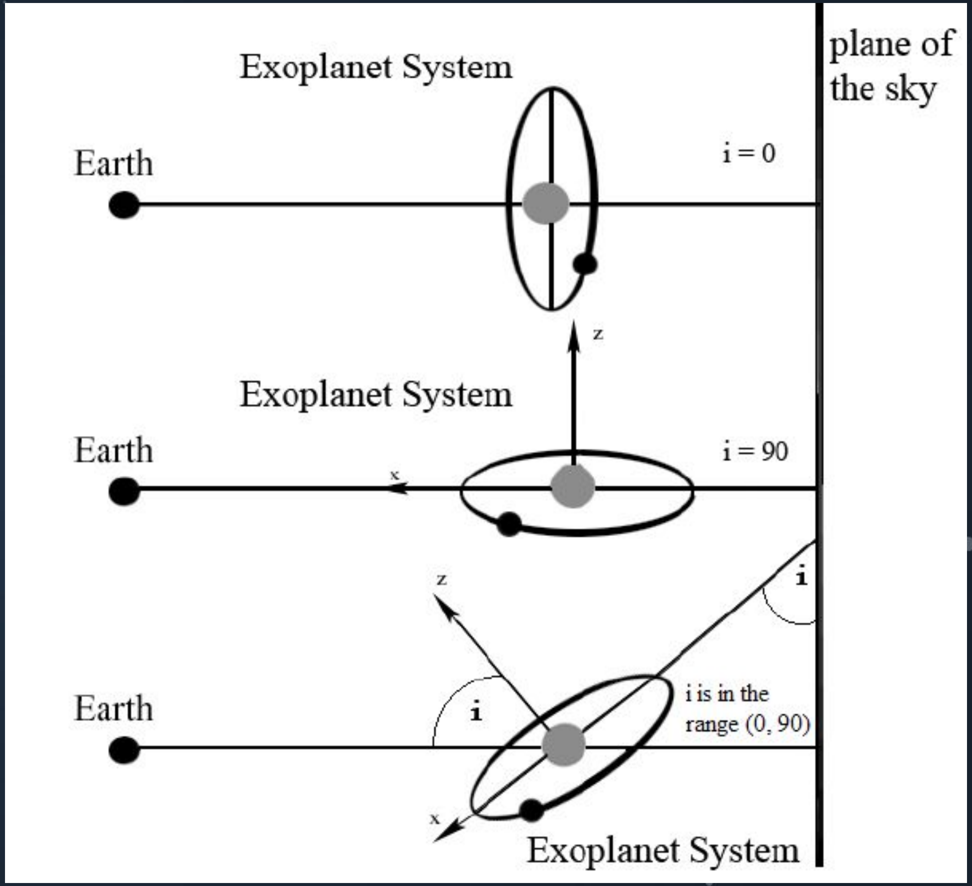
- ~1% for Jupiter-sized planets, ~0.01% for Earth-sized planets!
- Can measure planet's period, inclination, and radius



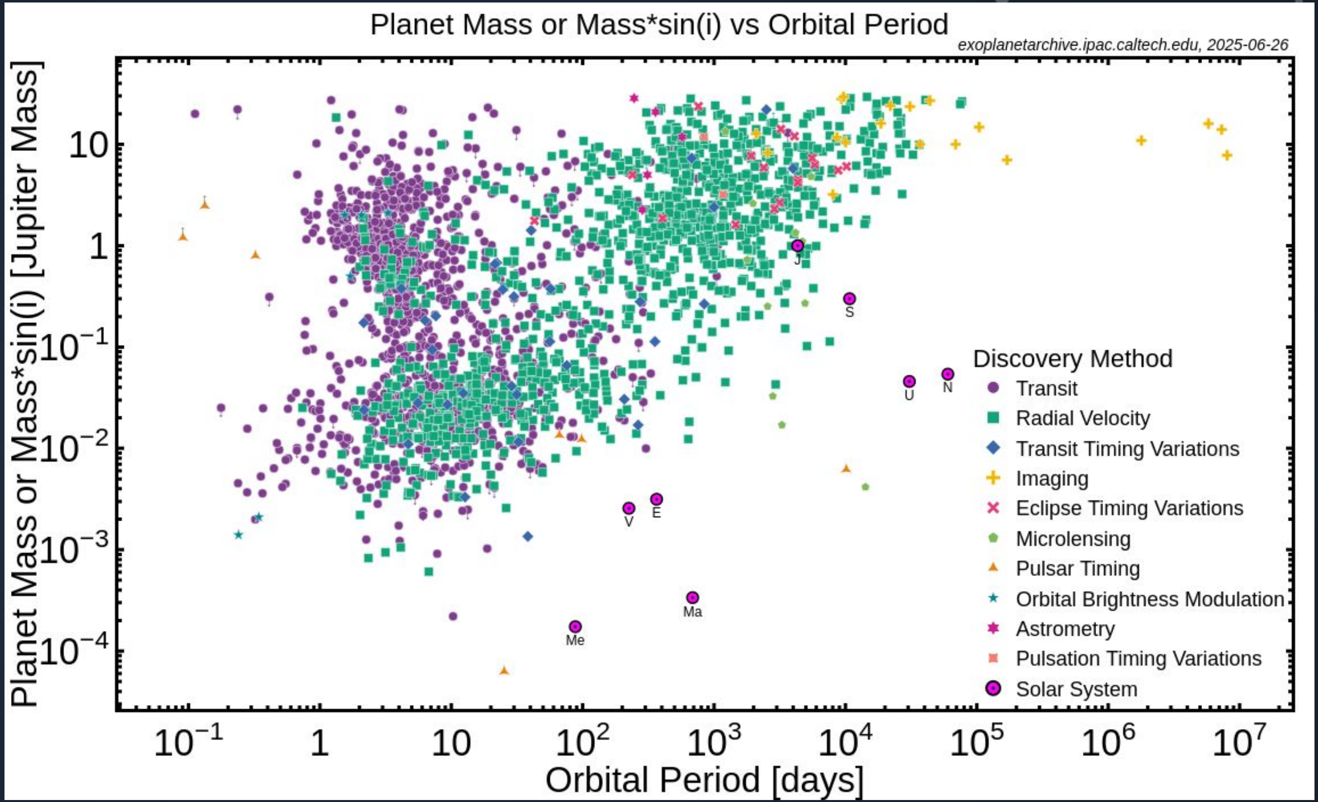
- Can only see planets that are *edge-on*!
- *Biased* towards finding bigger planets, since they produce larger transits
- Biased towards shorter-period planets, since they're more likely to transit

$$P_{tr} \sim R_*/a$$

(~0.5% for Earth)



- Possible to survey large parts of the sky at a time, to find many planets at once!
- Responsible for ~75% of planet detections

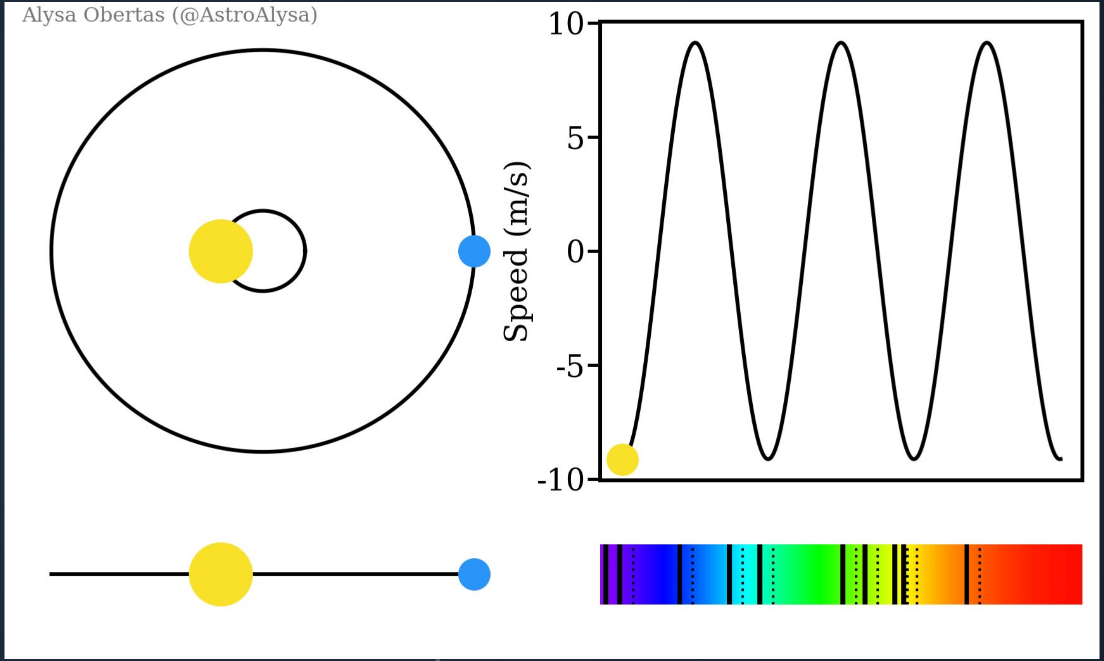


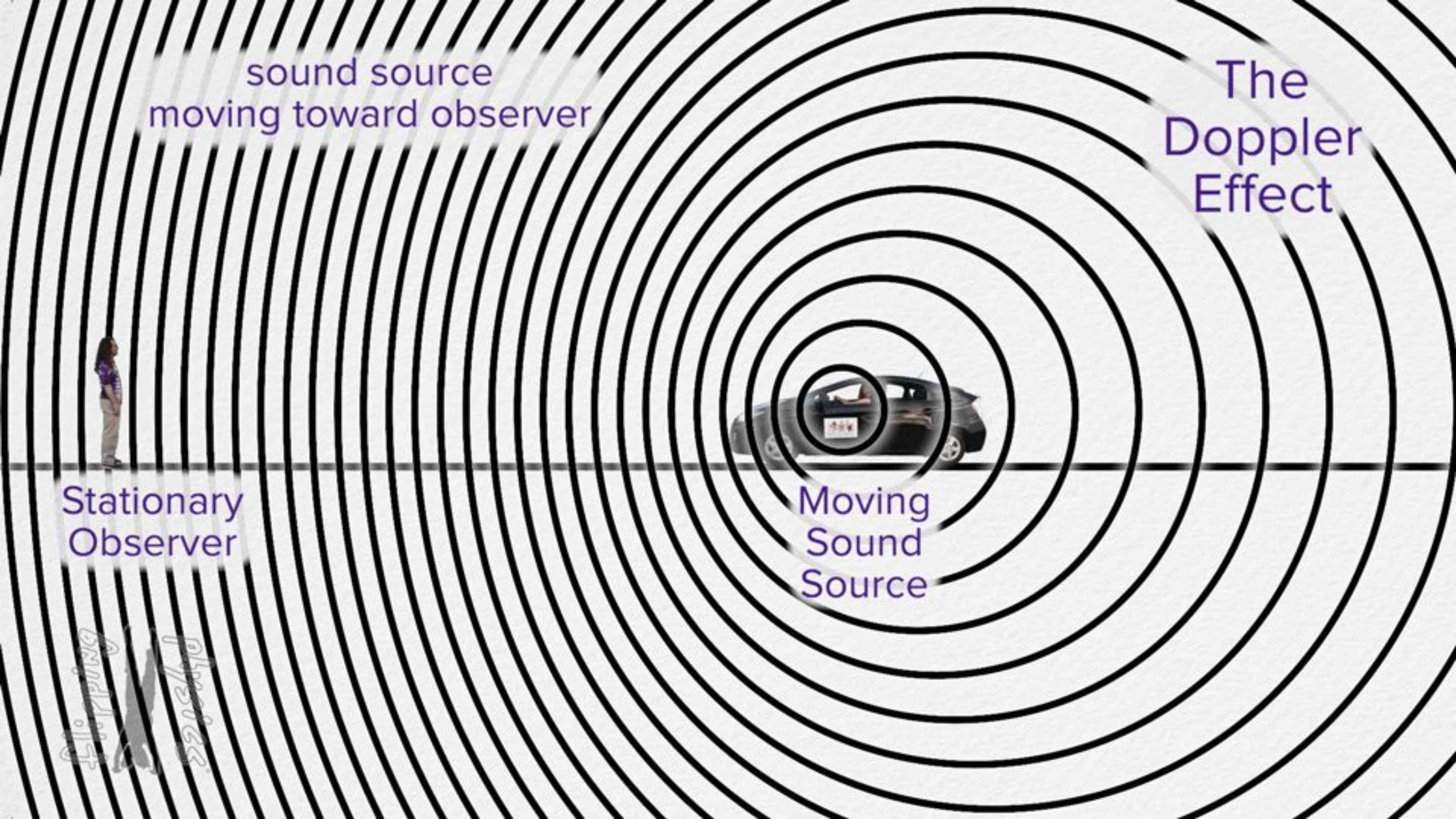
METHOD 2: RADIAL VELOCITIES (RVS)

- Time-resolved spectroscopy to measure a planet's **Doppler shift**

$$\lambda' = \lambda * \frac{\sqrt{1+v/c}}{\sqrt{1-v/c}}$$

- Can measure planet's period, eccentricity, and ***minimum* mass**




sound source
moving toward observer
The
Doppler
Effect
Stationary
Observer
Moving
Sound
Source

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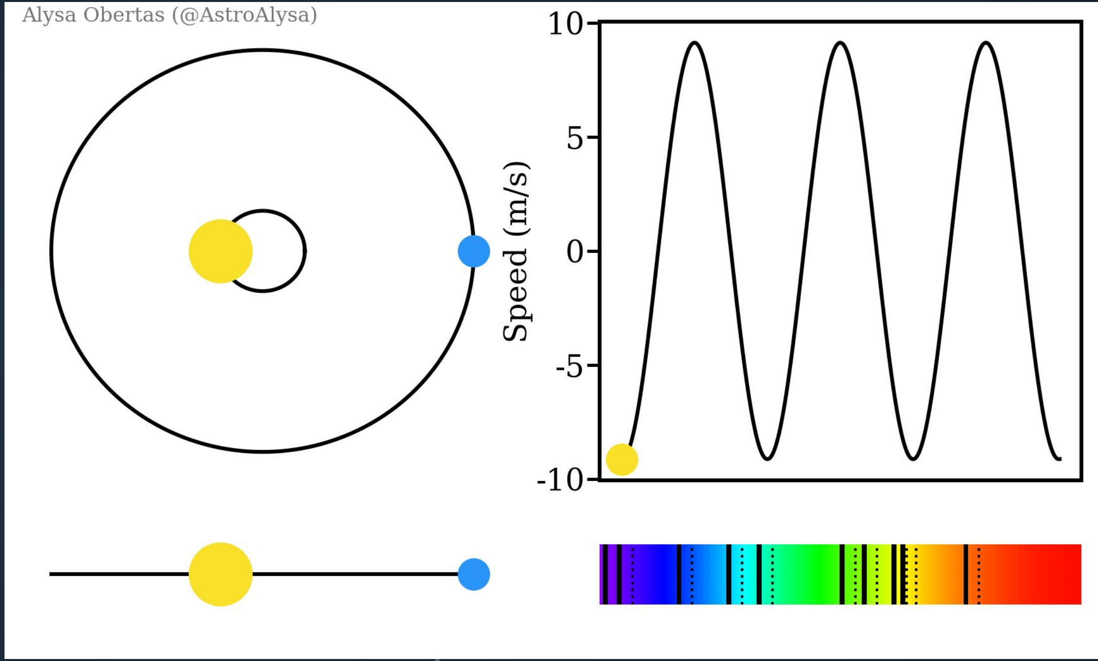
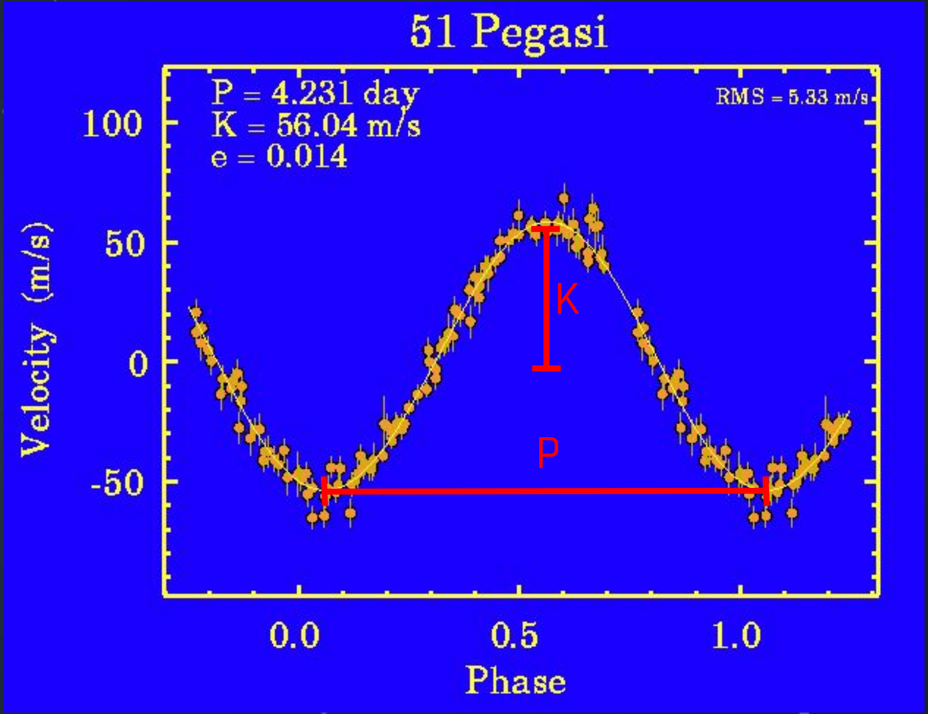
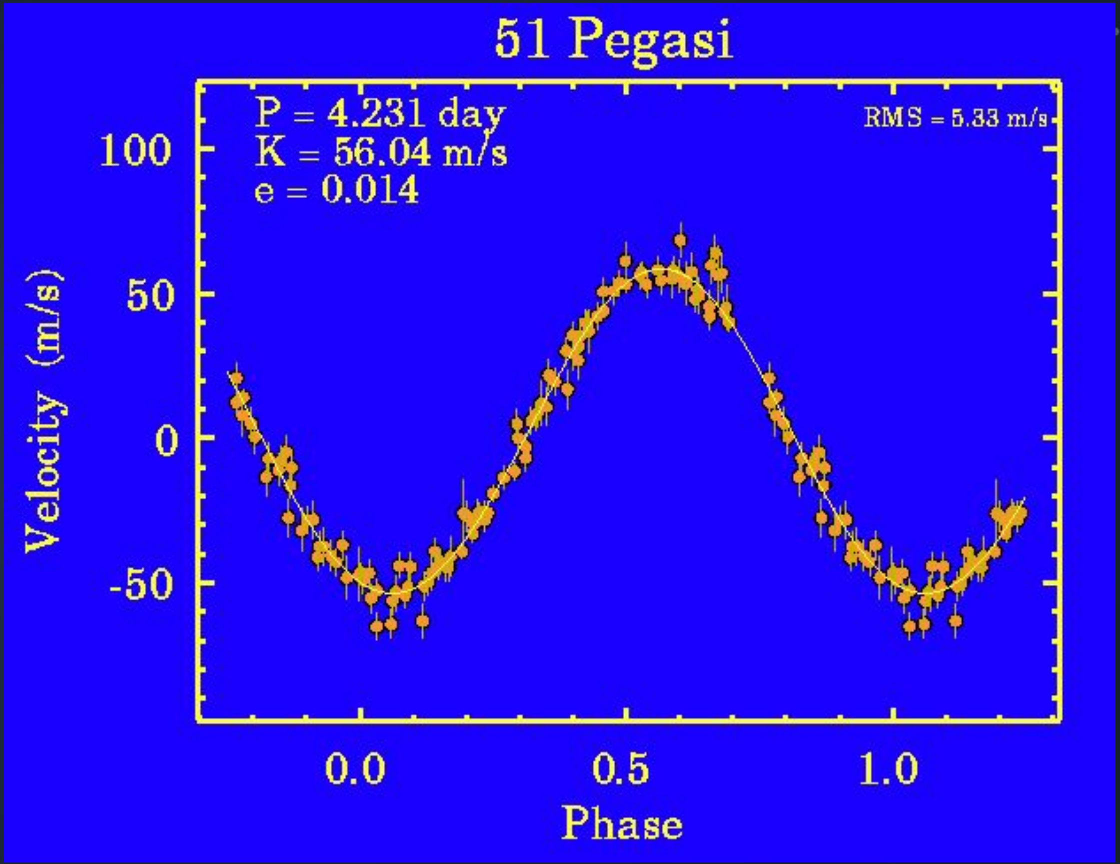


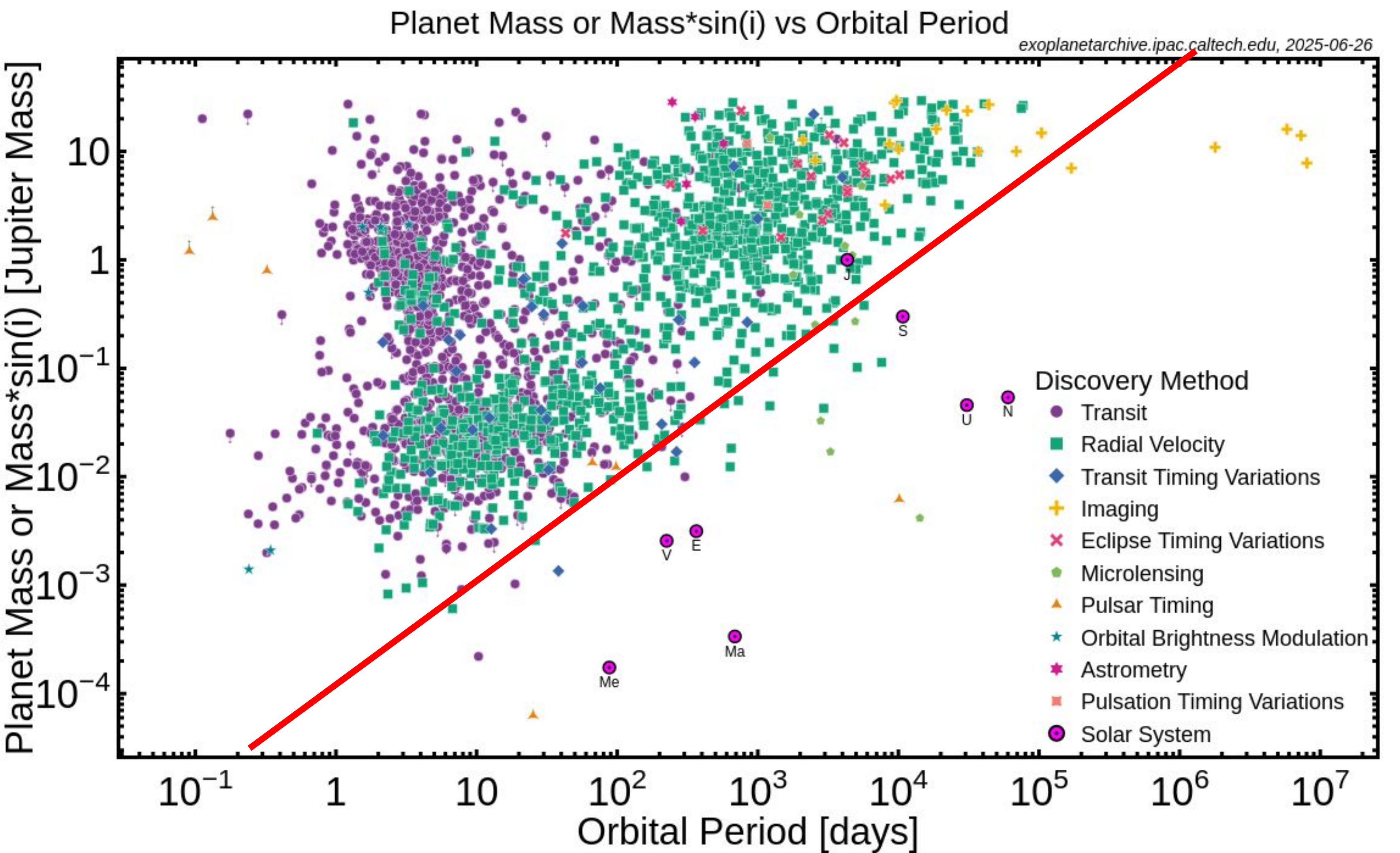
$$K = M_p \sin i \left(\frac{2\pi G}{P M_\star^2} \right)^{1/3}$$

- Biased towards bigger planets with shorter periods, and planets around smaller stars, since these produce bigger signals



- Best instruments in the world can reach precision of 1 m/s or lower!
- Can only do one star at a time, so less efficient than transits
- But planets don't need to be edge-on, so can see planets transits can't




Planet Mass or Mass*sin(i) vs Orbital Period
exoplanetarchive.ipac.caltech.edu, 2025-06-26
Planet Mass or Mass*sin(i) [Jupiter Mass]
Orbital Period [days]
Discovery Method
Transit
Radial Velocity
Transit Timing Variations
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Eclipse Timing Variations
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Astrometry
Pulsation Timing Variations
Solar System
Me
V
E
Ma
J
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N

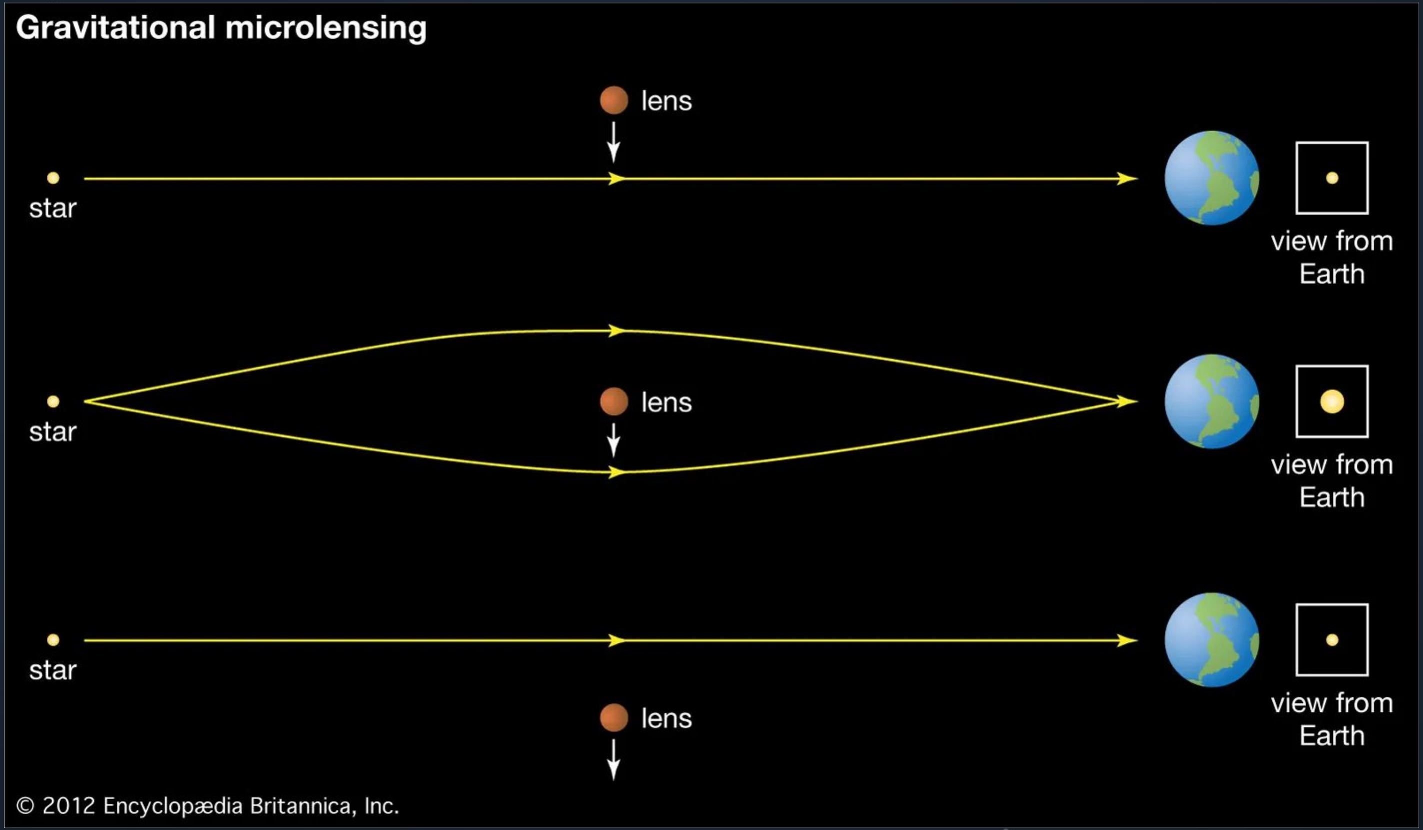
COMBINING TRANSITS AND RVS

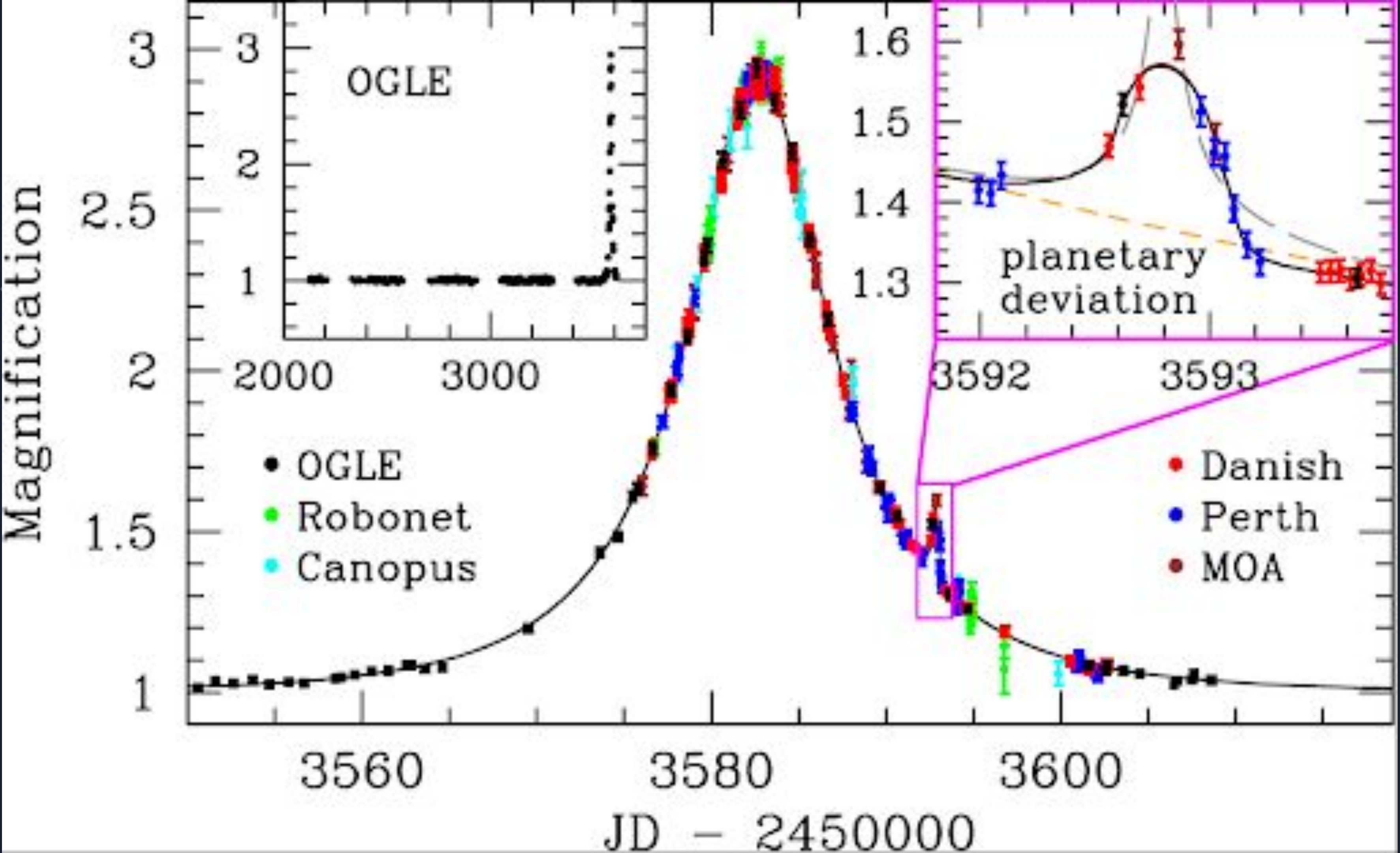
- Often useful to combine transits and RVs, they're complimentary!

- RVs tell you the mass, transits tell you about the radius -- combined, it can help you figure out a planet's composition!

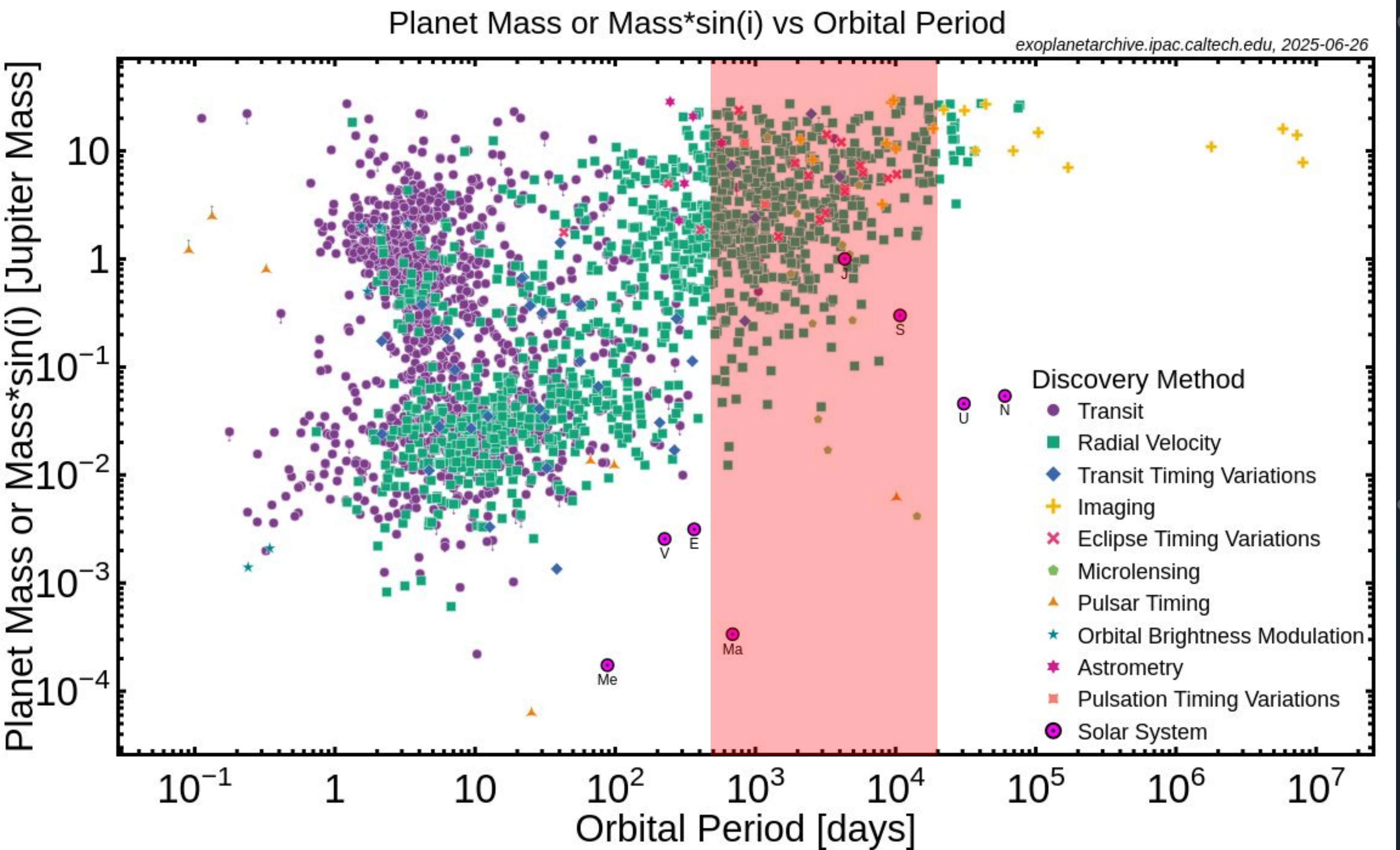
OTHER METHODS

METHOD 3: MICROLENSING





Magnification
3
2.5
2
1.5
1
OGLE
3
2
1
2000
3000
1.6
1.5
1.4
1.3
planetary deviation
3592
3593
OGLE
Robonet
Canopus
Danish
Perth
MOA
3560
3580
3600
JD − 2450000



Planet Mass or Mass*sin(i) vs Orbital Period
exoplanetarchive.ipac.caltech.edu, 2025-06-26
Planet Mass or Mass*sin(i) [Jupiter Mass]
Orbital Period [days]
10
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-1}
1
10
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
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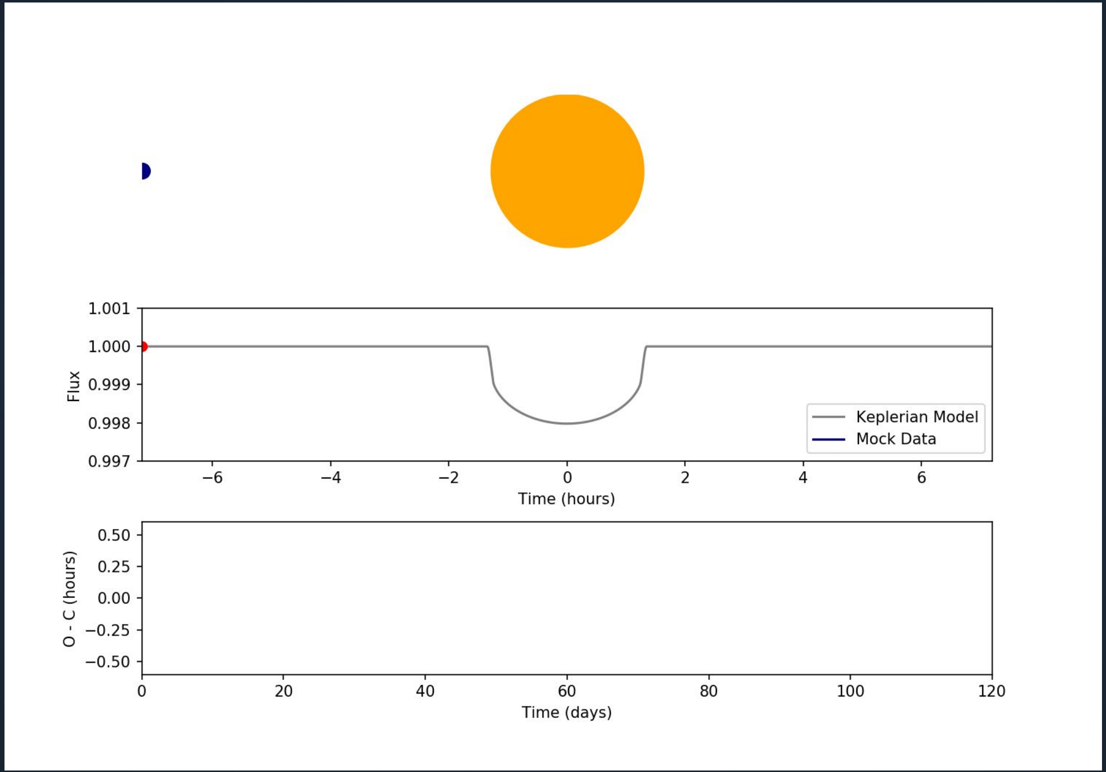
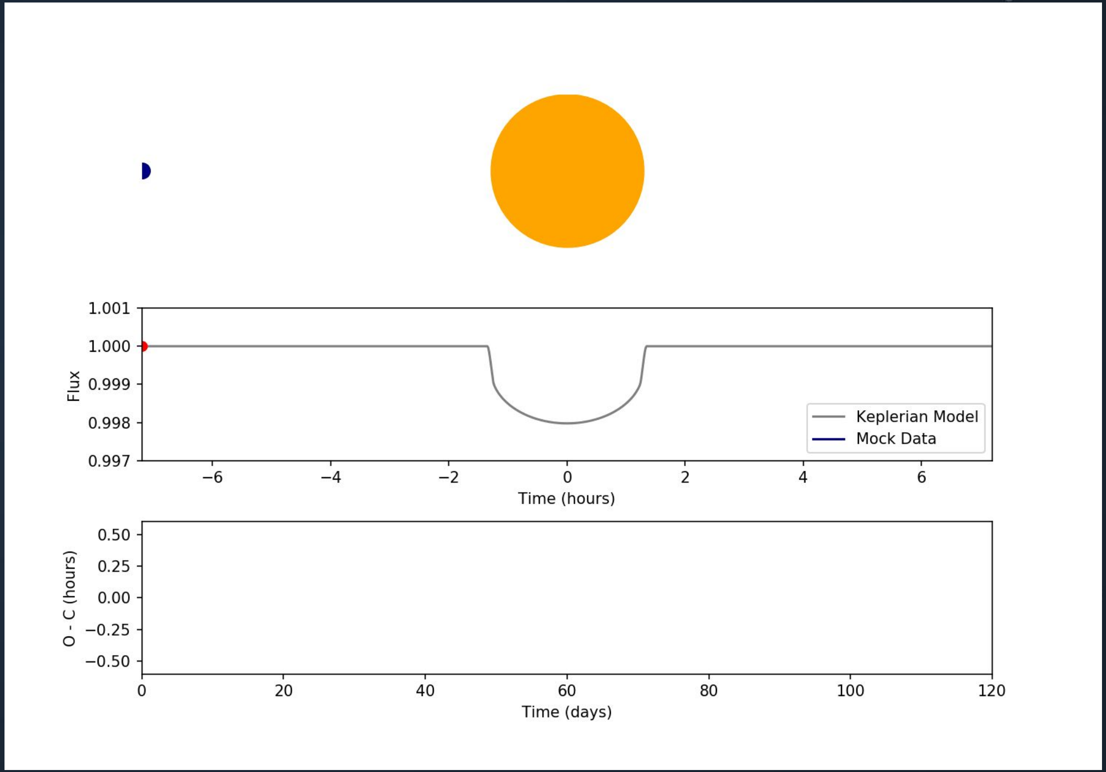
METHOD 4: TRANSIT TIMING VARIATIONS



Animations: Juliette Becker

METHOD 4: TRANSIT TIMING VARIATIONS

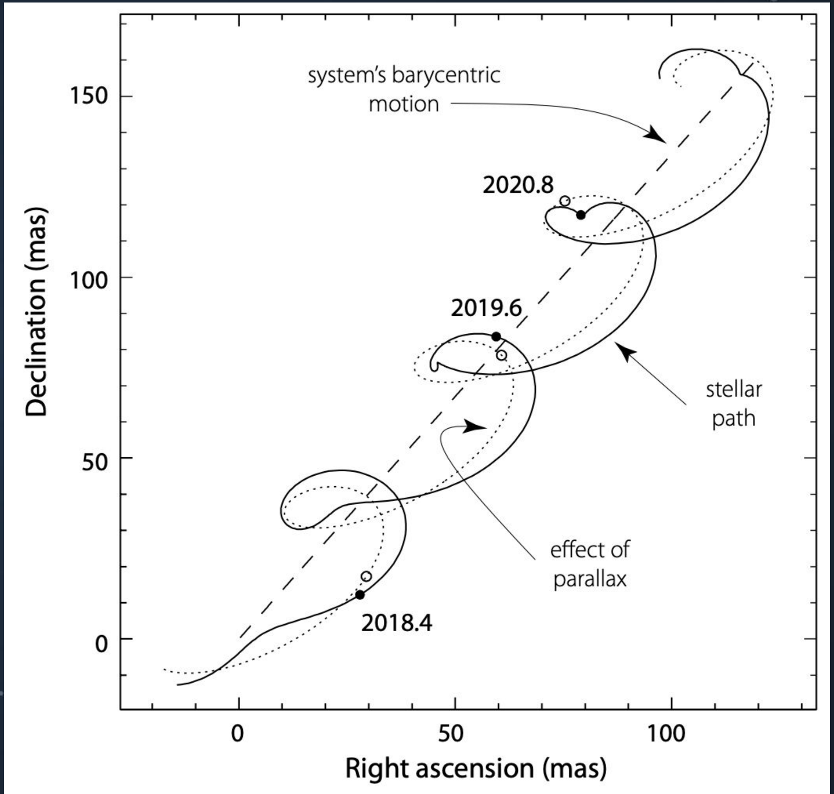
- Gravitational interactions with other planets can make planets transit minutes/hours early or late!
- Produce beat patterns of transiting early/late that we can use to reconstruct the unseen planet

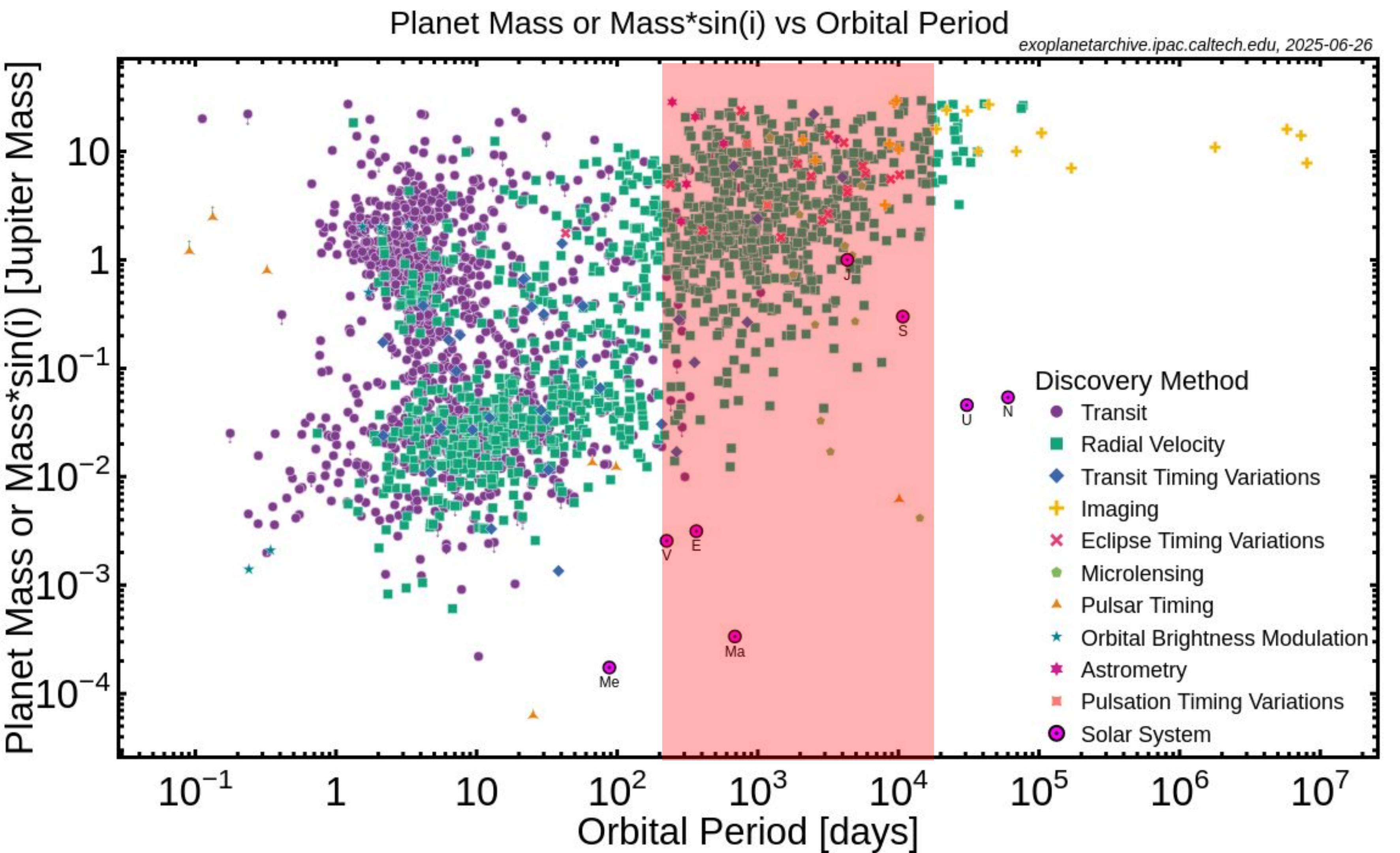


Animations: Juliette Becker

METHOD 5: ASTROMETRY

- Observe change in location of star due to planet motion
- REALLY hard to do, because stars barely move! Especially relative to motion of Earth around Sun or star's proper motion
- Final data from *Gaia* coming next year, should increase number of planets detected this way from 5 to ~many thousands
- Can tell period, eccentricity, minimum mass





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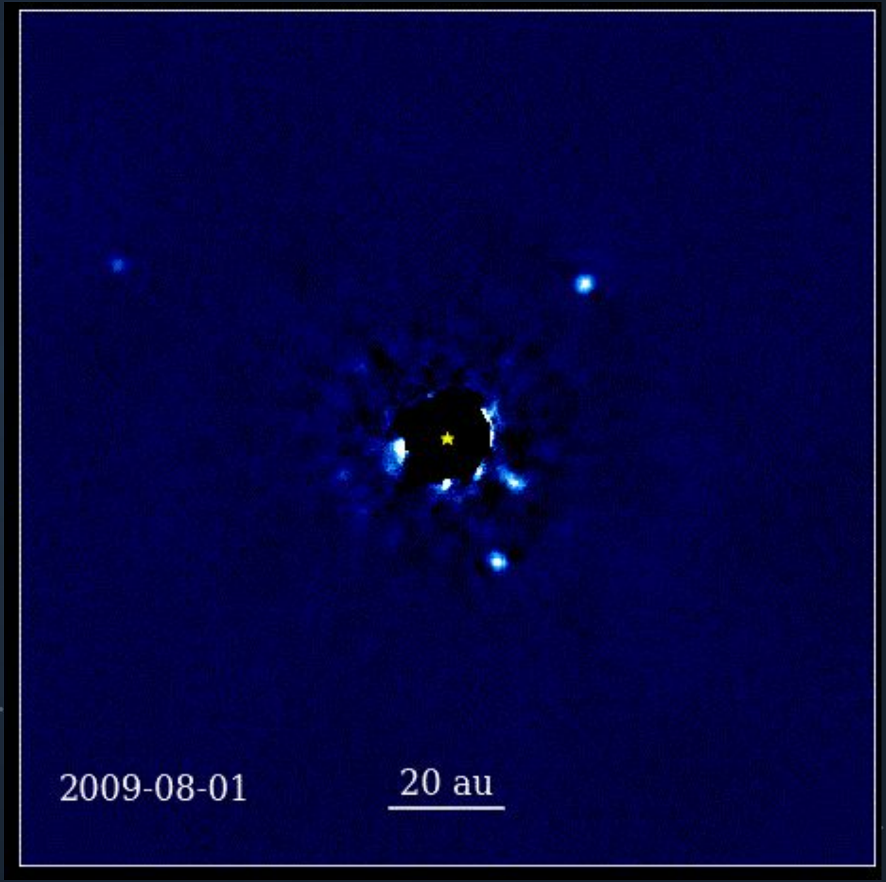
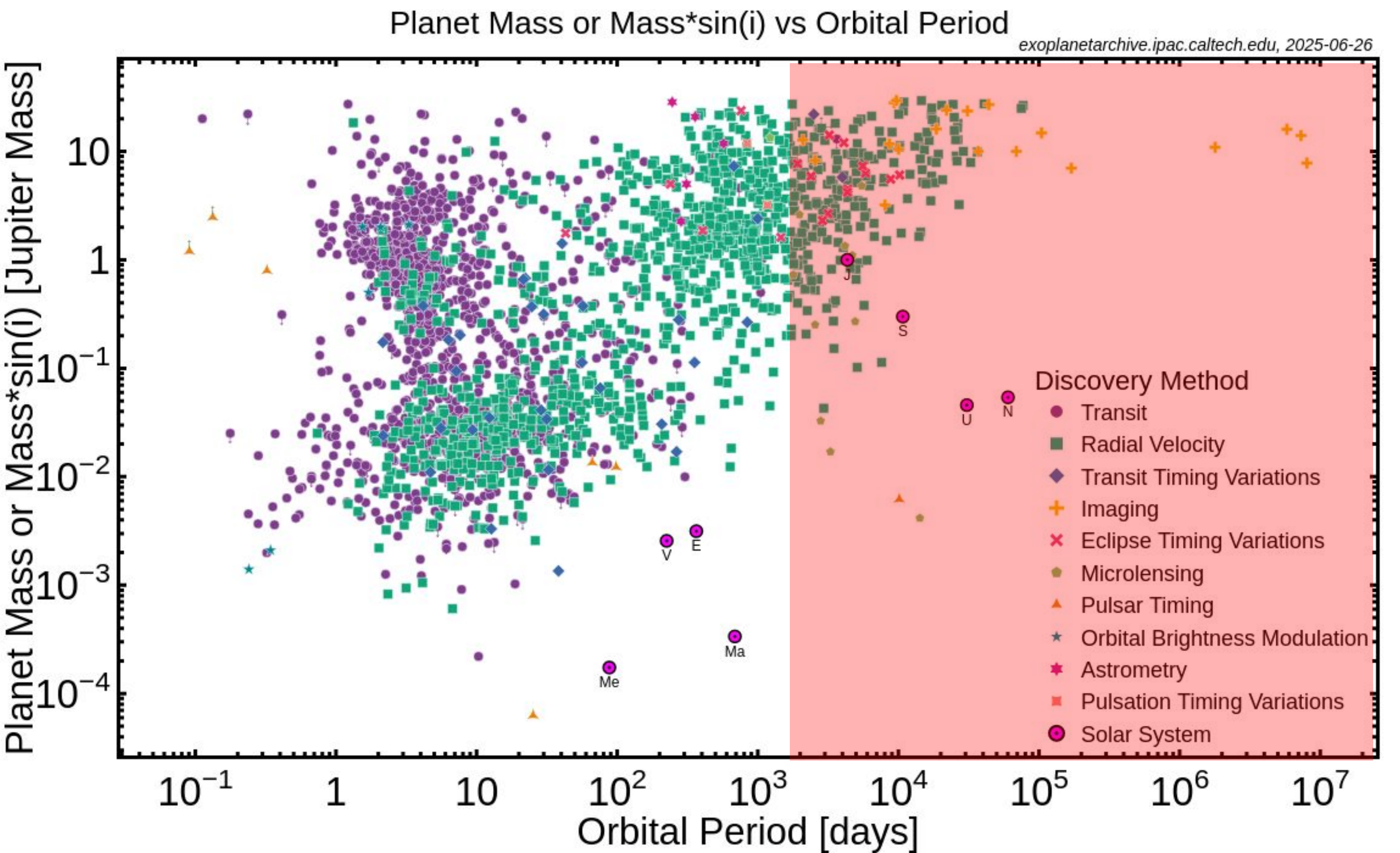
S
U
N

METHOD 6: DIRECT IMAGING

- See planets directly with big telescopes!
- Block out star with a **coronagraph**
- Only works on *big*, young planets really far away!
- Can tell planet period, eccentricity, rough idea of planet mass/size from brightness





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HOMEWORK

- Provided some simulated transit/radial velocity data
- Your goal is to fit it by hand!
- Optionally, fit it with computers and compare your results with the computer's